



Department of Education

STEM Education

National Schools of Excellence STEM Student's Resource Book Grades 11 Term 2

<u>DOMAIN</u>	<u>Phylum</u>
SCIENCE	Chemistry
Technology	
Engineering	
Mathematics	

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ISBN

Acknowledgements

This STEM Chemistry Student Resource Book for Grade 11 constitutes ten (10) Thematic Areas (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented.

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

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THEMA BIODIVERSITY

Module 1: Ethnobotany

Topic 1: Medicinal/ Herbal Plants

Activity: Mini Project

1. Document medicinal/herbal plants in Papua New Guinea.

Follow the documentation format provided by the teacher.

- Title.....
- Background.....
.....
- Methods.....
- Results/
Discussions.....
.....
.....
- Conclusion
.....
.....
.....

- Document traditional/herbal plants in your respective areas.
 - Identify a medicinal/herbal plant in their respective areas
 - Research, compile and document the particular plant
 - Keep an account of the documentation
 - Integrate in a system accessible for future reference to help Papua New Guinean citizens

Topic 2: Bioactive compounds from plants

Activity: Laboratory session

Title: Extraction of plant cyclotides

- In your groups divide yourself such that some students can harvest plant tissues while the others collect the target fish or larvae.
- Follow the format below to carry out your experiment and present your results and discussions.

Title: Extraction of Plant Cyclotides from _____ (Violaceae) or _____ (Fabaceae) and Bioassay against Test Targets

Objectives:

- To identify a local plant from the Violaceae/Fabaceae family
- To extract the plant materials using buffer
- To test the crude extract against _____ (a selected target)
- To observe and describe or measure the effects

Methodology:

- Test targets
 - Capture juvenile fishes (10-20 of them) or mosquito larvae (10-20 larvae)
 - Keep them in an aerated water tank
- Equipment
 - Domestic blender or mortar and pestle
 - Beaker (250mL) x 3
- Chemicals
 - Sodium phosphate (Na_4PO_4)
 - Ammonium chloride (NH_4Cl)
 - Sodium chloride (table salt)
- Buffer
 - Dissolve 50g of either sodium phosphate, ammonium chloride or sodium chloride in 500mL of water (either from the tap or distilled if available)
- Plant selection
 - Choose the plant of interest either from the Violaceae or Fabaceae families
 - Harvest the tissue of interest (for Violaceae the entire plant tissue can be used; for Fabaceae either the leaves, flowers or seeds can be used)
- Extraction
 - Extract 200g of plant tissues with buffer (250mL) in a blender. If there is no blender, grind the plant tissues in a mortar with pestle and then pour the ground material into the beaker and extract with buffer (250mL).

- **Bioassay**
 - Divide the test targets (fish or larvae) into two groups evenly
 - Keep them in a beaker of 100mL water
 - Add droplets of the crude extract and stir gently
 - Count the number of drops and observe the effect after each droplet
 - Continue adding the crude extract until there is no more changes in the behavior of the test targets
 - (Note that due to seasonal variations the effects may take some time and at higher concentrations of the crude extract)

Results and Discussions:

(Observe and discuss effects of the crude extracts on the test targets).

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Conclusions:

(Provide a succinct conclusion of the experiment).

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Reference:

(List the references you consulted for this laboratory session)

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Topic 3: Examples of Therapeutic Drugs

Activity: Case Study of Quinine and Artemisinin (Anti malarials)

1. Research on the therapeutic drugs, Quinine and Artemisinin and discuss how they were discovered;

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2. State their plant of origin and their pharmacological indication/use in your presentation.

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THEMA

BIOTECHNOLOGY

Module 1: Principles of Biotechnology

Topic 1: Polymerase Chain Reaction

Activity: Polymerase Chain Reaction

1. What is Polymerase Chain Reaction (PCR)?

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2. What are the Components of PCR?

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3. Briefly describe the PCR procedures?

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4. What are some useful applications of PCR?

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Topic 2: Dyes in DNA Sequencing

Activity:

Students are to;

1. Define the terms:

- Dyes.....
- DNA.....
- Gene.....
- DNA/Gene sequencing.....
- Genome.....
- Genome sequencing.....
- Describe the Sanger DNA sequencing method.

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Topic 3: Proteins

Activity Questions

1. What is a protein?

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2. How are Proteins synthesized?

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3. What are the functions of proteins?

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4. Case Study: Discuss Cry Proteins

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THEMA

HEALTH AND

MEDICINE

Module 1: Biochemistry in Medicine

Topic 1: How a Medical Drug Works (Mechanism of Action)

Activity

1. What are drugs?
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2. How are medical drugs classified?
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3. What are the purposes of medical drugs?
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4. How do drugs work?
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5. Briefly describe the mechanism of action of Aspirin and Non-steroidal anti-inflammatory drugs (NSAIDs)?
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6. How do NSAIDs get to the active site?
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7. What causes side effects in the case of aspirin? Why does aspirin cause stomach upset but the newer NSAID drugs do not?
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Topic 2: Drug Pharmacology

Activity

1. What is a drug receptor?

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2. Describe what efficacy and potency mean?

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3. Discuss First-Pass Effect?

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4. How do drugs cross membranes?

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5. Describe how drug metabolism and elimination occur.

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Topic 3: Significance of Diagnostic Laboratory Tests

Activity: Student Task

1. What is a full blood count (FBC) and why it is done?

- What are White blood cells;

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- Neutrophils?

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- Basophils?

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- Eosinophils?

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- What are Red blood cells?

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- What are Platelets?

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2. What is a liver function test? Briefly describe the role of each liver enzyme. Answer in terms of its significance.

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3. What is a kidney function test? Answer in terms of its measurable parameters and its significance.

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